

- I If L is regular language, then it is also DCFL and CFL.
 II Every DCFL is CFL, but it need not be regular.

3.6 Closure Properties of CFLs

Operation	Closed / Not Closed
Union ($L_1 \cup L_2$)	✓
Intersection ($L_1 \cap L_2$)	✗
Complement ($\bar{L}$)	✗
Set difference ($L_1 - L_2$)	✗
Concatenation ($L_1.L_2$)	✓
Reversal (L^{Rev})	✓
Kleene Closure (L^*)	✓
Kleene Plus (L^+)	✓
Subset (L)	✗
Prefix (L)	✓
Suffix (L)	✓
Substring (L)	✓
Substitution (L)	✓
Homomorphism (L)	✓
ϵ - free Homomorphism $h(L)$	✓
Inverse Homomorphism $h^{-1}(L)$	✓
quotient (L_1, L_2) = L_1/L_2	✗
Symmetric difference (L_1, L_2)	✗
Finite Union	✓
Finite Intersection	✗
Finite difference	✗
Finite concatenation	✓
Finite subset	✓
Finite substitution	✓
Infinite Union	✗
Infinite intersection	✗
Infinite difference	✗
Infinite concatenation	✗
Infinite Subset	✗
Infinite Substitution	✗
Union with Regular ($L \cup \text{Regular}$)	✓
$L \cup \text{Regular}$	✓
$L - \text{Reg}$	✓
$\text{Reg} - L$	✗

- Note :**(i) $CFL \cap CFL =$ Need not be CFL
- (ii) $CFL \cap \text{Regular} = CFL$ (Need not be DCFL)
- (iii) $CFL \cap DCFL =$ May or may not be CFL
- (iv) $CFL \cap \text{Finite} = \text{Finite}$
- (v) $CFL \cap \text{infinite} =$ Need not be CFL

3.7 Closure Properties of DCFLs

Operation	Closed / Not Closed
Union ($L_1 \cup L_2$)	×
Intersection ($L_1 \cap L_2$)	×
Complement ($\bar{L}$)	✓
Set difference ($L_1 - L_2$)	×
Concatenation ($L_1.L_2$)	×
Reversal (L^{Rev})	×
Kleene Closure (L) = L^*	×
Kleene Plus (L) = L^+	×
Subset (L)	×
Prefix (L)	✓
Suffix (L)	×
Substring (L)	×
Substitution (L)	×
Homomorphism (L)	×
$\in$ - free Homomorphism $h(L)$	×
Inverse Homomorphism $h^{-1}(L)$	✓
quotient (L_1, L_2)	×
Symmetric difference (L_1, L_2)	×
Finite Union	×
Finite Intersection	×
Finite difference	×
Finite concatenation	×
Finite subset	✓
Finite substitution	×
Infinite Union	×
Infinite intersection	×
Infinite difference	×
Infinite concatenation	×
Infinite Subset	×
Infinite Substitution	×
Union with Regular ($L \cup \text{Regular}$)	✓
$L \cup \text{Regular}$	✓
$L - \text{Regular}$	✓
$\text{Regular} - L$	✓

Note :

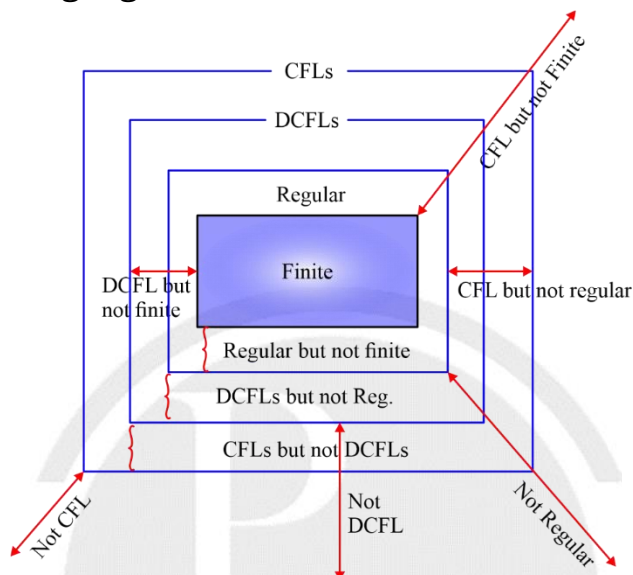
$DCFL \cup Regular$	: DCFL
$DCFL \cap Regular$	: DCFL
$DCFL - Regular$	: DCFL
$Regular - DCFL$	: DCFL
$DCFL \cup CFL$	: CFL (need not be DCFL)
$DCFL \cap CFL$	: Need not be CFL
$DCFL - CFL$	: Need not be CFL
$CFL - DCFL$	: Need not be CFL
$DCFL \cup Finite$	: DCFL
$DCFL \cap Finite$	: Finite
$DCFL - Finite$	: DCFL
$Finite - DCFL$	: Finite

3.8 Comparison of Regular Grammars and CFGs

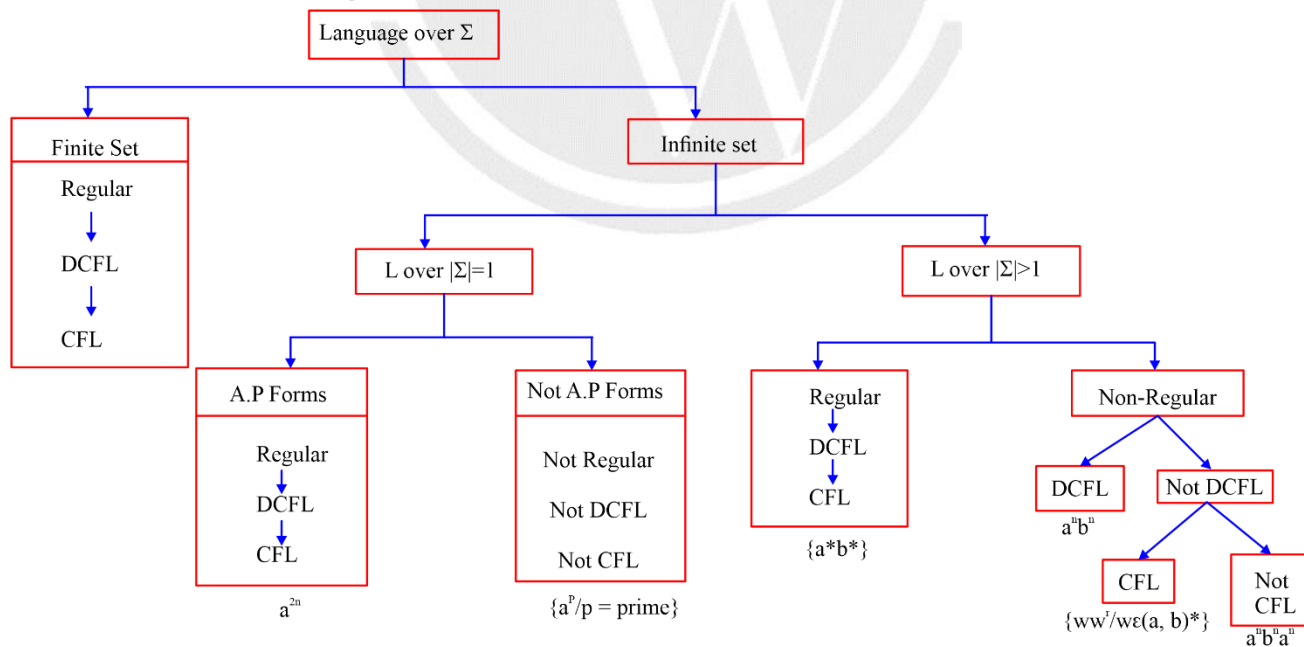
- CFL and CFG both are equivalent. $CFL \cong CFG$
- LLG: Left Linear Grammar, RLG: Right Linear Grammar, RG: Regular Grammar, LG: Linear Grammar, CFG: Context Free Grammar.
- Every LLG is RG
- Every RLG is RG
- Every RG is LG
- Every RG is CFG
- Every LG is CFG
- Every RG need not be RLG
- Every RG need not be LLG
- Every LG need not be RG
- Every CFG need not be RLG
- Every CFG need not be LG

3.9 Context Free Languages and DCFLs

I. Comparison of various languages



II. Identification of Regulars, DCFLs, and CFLs



- [1] $L = \{a^n b^n c^k \mid n, m, k \geq 0\}$
 $= a^* b^* c^*$
 $= \text{Regular}$
- [2] $L = \{a^n b^n \mid n \geq 0\}$
 $L = \text{DCFL but not regular}$
 $S \rightarrow aSb \mid \epsilon$
- [3] $L = \{a^n b^{2n} \mid n \geq 0\}$
 $L = \text{DCFL but not regular}$
 $S \rightarrow aSbb \mid \epsilon$
- [4] $L = \{a^{2^n} b^n \mid n \geq 0\}$
 $L = \text{DCFL}$
 $S \rightarrow aaSb \mid \epsilon$
- [5] $L = \{a^{2^n} b^{2^n} \mid n \geq 0\}$
 $L = \text{DCFL}$
 $S \rightarrow aaSbb \mid \epsilon$
- [6] $L = \{a^m b^n c^* \mid \text{Assume always } n \geq 0 \text{ in all examples}\}$
OR
 $= \{a^m b^n c^* \mid m = n\}$
 $\text{DCFL but not regular}$
- [7] $L = \{a^n b^* c^n\}$ DCFL
- [8] $L = \{a^* b^n c^n\}$ DCFL
- [9] $L = \{a^m b^n c^* \mid m \neq n\}$
DCFL
- [10] $L = \{a^m b^n c^* \mid m < n\}$
DCFL
- [11] $L = \{a^m b^n c^* \mid m > n\}$
DCFL
- [12] $L = \{a^m b^n c^* \mid c \leq n\}$
DCFL
- [13] $L = \{a^m b^n c^* \mid m \geq n\}$
DCFL
- [14] $L = \{a^m b^n c^{m+n} \mid m, n \geq 0\}$ is DCFL



[15] $L = \{ a^{m+n}b^{n+k}c^{k+m} \mid m, n \geq 0 \}$ is CFL

[16] $L = \{ ww^r \mid w \text{ belongs to } \{a, b\}^* \}$ is CFL but not DCFL

